

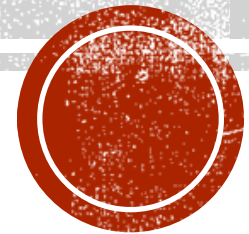


CS 517: Digital Game Design

Game Research Talk

ENHANCING COMMUNICATION AND SOCIAL INTEGRATION IN AUTISM SPECTRUM DISORDER THROUGH SERIOUS GAMES

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Outline

- Overview of ASD and its challenges.
- Role and impact of serious games in educational settings.
- Each paper's contributions, methodologies, and outcomes.
- Comparative analysis and implications
- Future developments

Introduction

- Exploring Game Design and Empathy for Neurodivergent people.
- This literature review explores the integration of role-playing and interactive gaming elements into serious games aimed at enhancing empathy and understanding towards individuals with Autism Spectrum Disorder (ASD).
- Three websites as a searching source,
 - Google Scholar
 - ACM
 - IEEE Explore

Autism Spectrum Disorder

- ASD is a complex developmental condition that involves persistent challenges in social interaction, speech and nonverbal communication, and repetitive behaviors.
- **Communication Barriers**
 - Struggle with understanding and using verbal and nonverbal cues effectively.
- **Social Challenges**
 - Difficulties in social awareness and interactions can lead to social isolation and emotional distress.
- **Family Dynamics**
 - Families often need significant support to navigate the challenges of raising a child with ASD, including specialized care and education.

Key Characteristics of Autism Spectrum Disorder

- Difficulty with social cues, body language, and understanding unspoken rules.
- Preference for routines, discomfort with changes, repetitive behaviors for comfort
- Strengths in areas like memory, math, art, music, alongside areas that may need support.



Paper 1

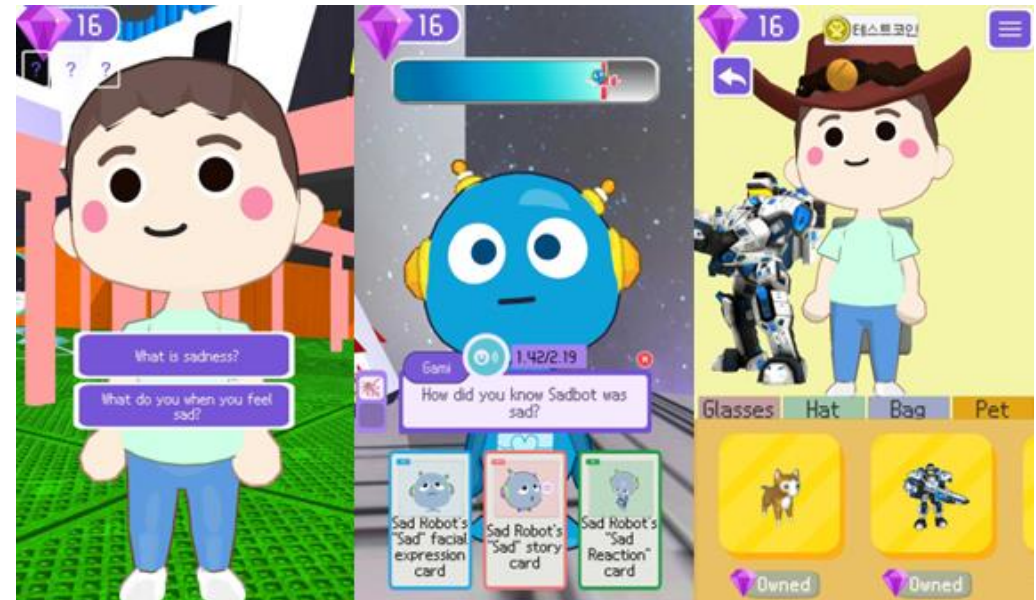
Game-Driven Practices for Social Skills: Exploring Usability for Children with Autism Spectrum Disorder

■ Source:

ACM 3rd International Conference on Information Technology for Social Good (GoodIT 2023)

■ Authors:

- Bryan Kim, Rensselaer Polytechnic Institute
- Da-Yea Song, Seoul National University
- Doori Kim, Konkuk University & MeDKit Inc.
- Ye Chan Kim, MeDKit Inc
- Ill Jin Lee, Konkuk University
- Hee Jeong Yoo, Seoul National University
- Jin Joo Son, MeDKit Inc.
- Jae Hyun Han, Seoul National University
- Jee-In Kim, Konkuk University & MeDKit Inc.



■ Research Area:

Serious games for developmental disorders, focusing on innovative methods to enhance social skills in children with Autism Spectrum Disorder (ASD).

Paper 1

Game-Driven Practices for Social Skills: Exploring Usability for Children with Autism Spectrum Disorder (Continued)

- **Contribution:**

1. Development of emotion adventure to teach social cognition skills to children with ASD
2. Effective engagement of children in activities that enhance their ability to respond to social cues
3. Evaluates the usability and accessibility of game's interface

- **Key Findings:**

- Positive reception from children, indicating a high level of engagement and improvement in recognizing social cues.
- Potential identification of biomarkers for ASD interventions through gameplay performance analysis.

- **Methods:**

1. In-depth development process to ensure the game meets the specific needs of children with ASD.
2. Collection and analysis of gameplay data to assess user engagement and educational impact.

Paper 2

Serious Games to Facilitate Social Integration for Children with Autism Spectrum Disorder

■ Source:

ACM 23rd International Conference on Human-Computer Interaction (Interaccion 2023)

■ Authors:

- Javier Jiménez-Honrado, Instituto de Desarrollo Industrial, Spain
- Jose A. Gallud, Universidad de Castilla-La Mancha, Spain
- Gabriel Sebastián, Universidad de Castilla-La Mancha
- Víctor M. R. Penichet, Universidad de Castilla-La Mancha



■ Research Area:

Focuses on the application of serious games for enhancing the social integration of children with Autism Spectrum Disorder (ASD), using technology to tailor educational experiences

Paper 2

Serious Games to Facilitate Social Integration for Children with Autism Spectrum Disorder (Continued)

- **Contribution:**

1. Innovative educational tools in the form of mini-games
2. Personalized learning approaches to the unique needs of children with ASD
3. Promotion of social skills in a structured yet engaging manner

- **Key Findings:**

- Children with ASD show improved social integration skills through interactive and engaging gameplay.
- High adaptability of game settings to meet diverse learning needs and preferences of children with ASD.

- **Methods:**

1. Detailed account of the design and implementation of educational mini-games within a serious games framework.
2. Integration of continuous feedback loops to assess game effectiveness and user engagement.

Paper 3

An Interactive Serious Game to Target Perspective Taking Skills Among Children with ASD: A Usability Testing

- **Source:**

Journal of Behavior and Information Technology, Taylor & Francis (2021)

- **Authors:**

- Parisa Ghanouni, Dalhousie University, Canada
- Tal Jarus, University of British Columbia, Canada
- Jill G. Zwicker, University of British Columbia, Canada
- Joseph Lucyshyn, University of British Columbia, Canada

- **Research Area:**

Examines the use of interactive motion gaming programs to improve perspective taking in children with ASD, focusing on user-centered development and stakeholder involvement in the design process.

Paper 3

An Interactive Serious Game to Target Perspective Taking Skills Among Children with ASD: A Usability Testing (Continued)

■ Contribution:

1. Integrated feedback from children with high-functioning ASD and their parents
2. Emphasized the importance of design, user engagement, and fine-tuned feedback based on direct input from primary users and their caregivers.

■ Key Findings:

- Demonstrated the value of incorporating direct feedback from children and parents in the iterative design process to ensure the game meets user needs and preferences.
- Adjustments made to game visuals, animations, and feedback mechanisms to better cater to the sensory sensitivities and interaction preferences of children with ASD.

■ Methods:

1. Engaged children with ASD and their parents in the design and testing phases to gather essential feedback for game refinement.
2. Employed methods like 'think aloud' during game trials to capture real-time reactions and suggestions for improvement from users.

Comparing Contributions

- **Progression in Interaction Complexity**

From structured digital role-playing (Paper 1) to varied mini-games (Paper 2), culminating in interactive motion gaming (Paper 3).

- **Stakeholder Involvement**

Increases from user feedback post-development in Papers 1 and 2, to extensive pre-development involvement and testing in Paper 3.

- **Technological Integration**

Evolves from simple game mechanics to sophisticated motion-sensing technology, reflecting greater technological integration and user interaction fidelity.

Applications To 'Running Household Chores With Samuel'

▪ **Paper 1: Game-Driven Practices for Social Skills**

- Incorporate role-playing elements where neurotypical players assume the roles of neurodivergent individuals.
- This can help players understand the challenges faced by Samuel during household chores, promoting empathy and awareness.

▪ **Paper 2: Serious Games to Facilitate Social Integration**

- Develop specific tasks such as organizing, cleaning, or making foods. Each task can highlight different aspects of Samuel's approach to chores
- Underscoring the need for structure and predictability often preferred by neurodivergent individuals.

▪ **Paper 3: Interactive Serious Game to Target Perspective Taking Skills**

- Implement interactive elements where players must physically instruct or react to Samuel's actions via voice command or similar technology.
- This can deepen understanding of Samuel's physical movements and sensory experiences during chores, enhancing neurotypical players' perspective-taking capabilities.

Future Developments

- Expanding Game Scenarios
- Advanced Interaction Techniques
- Adaptive Game Mechanics
- Longitudinal Impact Studies
- Collaborative Gaming
- Inclusive Game Development

Conclusion

- Each game component educates players about the sensory and cognitive processing differences
- That will affect how neurodivergent children like Samuel perceive and engage with their environment.
- By actively participating in Samuel's household chores through varied interactive methods inspired by these papers, players can gain understanding of Samuel's experiences and challenges.

THANK YOU